SCREAMINGFACE · URL4 · DESIGN DECISION

url4 topology — endpoint, node, discovery, addressing, transport

Sharp definitions before the Engine grows further

STATUS	proposed — owner review; one decision left open in §5; aligned with the url4-refactor drafts 2026-09-08 (§12); vocabulary realigned to the spec's Node/Endpoint
WORK ITEM	OME-1110
DATE	2026-09-04
OWNER	Sergey Bershadsky (execution and telemetry architecture)
GRAMMAR OWNER	Kevin McDonough (URL4 spec, Parts A/B)
SPEC POINTERS	URL4.ai Specification v0.5 DRAFT — Parts A/B (main) and Parts C–I drafts (url4-refactor branch)

Read §0 first. Everything else is the reasoning behind one line of that table.
Open questions and spec deltas live in the companion, open-work PDF.

0. The eight answers

This document fixes the words we use for the pieces of url4. It is short on purpose. Each section states a definition, says where the spec agrees or is silent, and stops.

#	QUESTION	ANSWER
1	What is an endpoint? What is a node?	An endpoint is a stateless function behind a path (`/claude`): it takes `(context)!intent` , returns a result, and can evaluate a full url4 expression. A node is the origin that serves a set of endpoints, `/` being its default.
2	Who owns what?	The node owns discovery, credentials and outbound traffic (§4, §11). Endpoints own nothing but their function. Every requester that runs an evaluator is itself a node (§4, §13).
3	Swarm? Composition?	Not protocol words. The expression **is** the composition. The nodes it touches are its request tree. “Swarm” is only a deployment word for one operator’s nodes.
4	`.well-known` or OPTIONS?	Three mechanisms now: the node document (Part G §27.1), the `Capabilities` response header (also §27.1), and OPTIONS (ours). §5 documents all three; the owner picks there.
5	`/name` vs `url4://name` ?	`/name` is an endpoint on the node that is evaluating. `url4://name` is node `name` ’s default endpoint `/` . Spec Part B §5.4 already says this.
6	Where does an ensemble run?	Wherever an evaluator runs: the SDK on a laptop, or a node. A local node may mount remote endpoints under local names (proxy mounts).
7	Can I test a node first?	One fetch of the node’s capabilities document, or one OPTIONS per endpoint (§5). A dry-run **plan** on the node is reopened as a question (§10).
8	Streaming fallback?	One GET asks for the richest mode; the endpoint answers with the best it has: WebSocket, then SSE, then sync (§6). Sync is the only MUST. Async is the spec’s third `delivery` value.
9	Text in, text out?	A habit, not a rule. Every edge carries a **typed payload** (text, image, audio, video, embeddings) named by its media type. Text is the default. No grammar change (§8).

What changes against today

- The Engine becomes a url4 **node**. Today no url4 endpoint is reachable over HTTP; the SDK’s endpoint-to-endpoint code is unused.
- Every endpoint speaks the same GET. The endpoint picks the richest delivery it supports: **WebSocket → SSE → sync**, in one round trip. Sync is the only MUST.
- We keep the spec’s two words: a **node** is an origin that serves a set of **endpoints** (`/claude` , `/codex` , …). No rename is asked of Kevin; “host” is gone from this document.
- An endpoint is a **universal inference processor**: image generation, speech, video and multimodal ensembles become ordinary expressions on the same engine, telemetry and cost accounting (§8).

1. Terms

Eight words. Each has one meaning. The first two are the spec’s own, unchanged.

OUR WORD	MEANING	SPEC WORD TODAY	IN THE SDK	IN THE ENGINE
Endpoint	A stateless function at a path. Accepts `GET <path>?q=<expr>` , evaluates it, returns a result plus envelope.	Endpoint (Part A §1.4), same word	an entry in `Url4Node` ’s endpoint registry	a route such as `/anthropic/<model>`

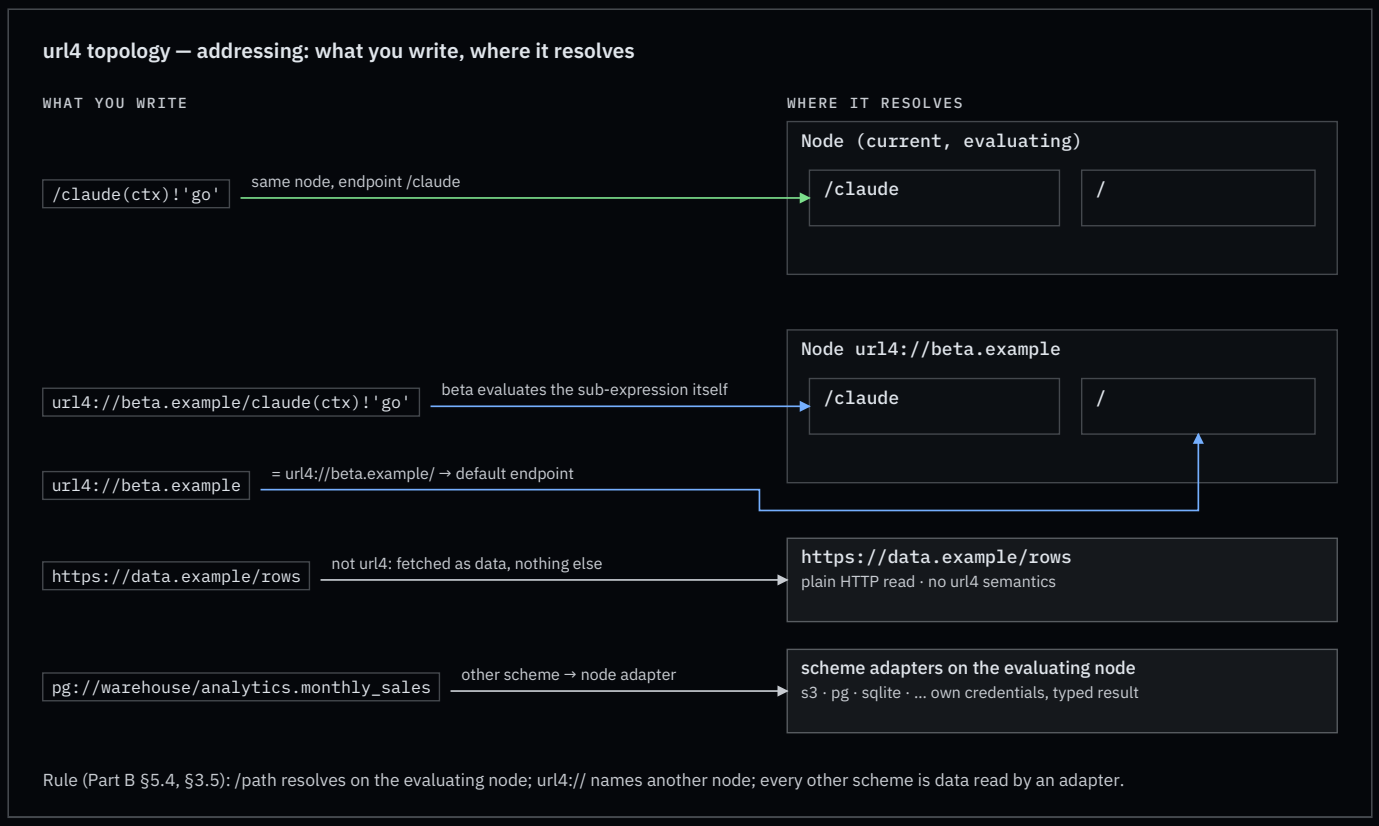
OUR WORD	MEANING	SPEC WORD TODAY	IN THE SDK	IN THE ENGINE
Node	An origin that serves a set of endpoints, `/` being the default, and answers discovery.	Node (Part A §1.4), same word	`Url4Node`	the App, after this change
Mount	The binding of a path to an endpoint implementation: `local` (in-process), `command` (subprocess), or `proxy` (forwards to a declared remote endpoint).	not defined	`endpoint()`, `[commands]`, —	in-process handlers only
Evaluator	Whatever runs an expression: resolves sources, fans out, reduces, runs the intent. Every endpoint contains one.	"the endpoint executes" (Part A §1)	`url4.dag.run`	`Url4Executor` in the Runner
Intent processor	The thing inside an endpoint that turns resolved context plus intent into a result: a model call, a script, a command.	Intent processor (Part A §1.4)	endpoint handler	one `httpx` call to aigateway
Requestor	Whoever sends an expression.	Requestor (Part A §1.4)	`Client`	product client
Request tree	The nodes and endpoints one expression touches. Strict tree, never a graph (Part H §29.1).	"request tree", "call tree" (Part H §31)	the compiled DAG, per endpoint	one run
Capabilities document	JSON a node publishes to say which endpoints, mounts, features and delivery modes it offers.	Part G §27.2 (draft; single endpoint, `collections`, `intent_processors`)	none	none

Words we do not use in the protocol: ensembler, orchestrator, swarm, composition, plan, fusion. "Fusion" stays product copy.

Naming note: Parts C–I on the `url4-refactor` branch still use the older `abc://` scheme and `ABC-*` headers; Parts A/B use `url4://`. This document writes `url4` and `URL4-*` throughout and assumes the rename Kevin lists as his open question 19 (Part I §43).

2. Addressing

Three address forms, plus any other scheme as data. The spec fixes their meaning in Part B §3.1.1, §3.5 and §5.4; we add one rule about the root and one about other schemes.



- `/claude(ctx)!'go'` — an endpoint on the node that is evaluating. Resolves to `url4://<current node>/claude` (Part B §5.4).
- `url4://beta.example/claude(ctx)!'go'` — an endpoint on another node. That node evaluates the sub-expression itself (Part B §3.1.1; SDK invariant OME-535).
- `url4://beta.example` — the node's default endpoint. **Our rule:** `url4://node` ≡ `url4://node/`.
- `https://data.example/rows` — plain HTTP data. No url4 headers, no envelope, `@` is an error (Part B §3.5).

Root and version. `/` is the node's default endpoint. The spec makes `/v1` the protocol-version path (Part D §19.1) but its own examples already write `abc://thepost?q=…` with no path at all (Part I §41.27). We keep both: `/` serves the current version; `/v1` is an explicit alias. Two tensions to settle with Kevin: Part D §19.4 gives the `v` parameter precedence over the path, and §19.3 says a nested expression takes its version from the path it targets, which a bare node address does not carry. The SDK's `url4 serve` (`/v1`) and `Client` (default `/v1`) move to `/` (Appendix B).

Scheme inheritance. A bare relative data URI inherits the scheme of the request that carried it (Part B §5.4.1). Under `url4://` it is a url4 read; under `https://` it is a plain read.

Any scheme is a source. The spec fixes the roles of `url4://` (evaluate) and `https://` (read), lists `s3://` with the rule "the endpoint uses its own credentials", and fails unknown schemes with `unsupported_mode` (Part B §3.5). We generalise the `s3://` rule: any other scheme is a **read through a scheme adapter that the evaluating node mounts**. The SDK already has the slot: `FetchRequest.kind` is `url4`, `http`, `relative`, or `other`.

```
(sales=pg://warehouse/analytics.monthly_sales,
 logo=s3://brand-assets/logo.png;accept=image/png,
 notes=sqlite:///srv/app/notes.db/notes,
 policy=https://data.example/policy.md)!'Draft the Q3 update. Use the logo.'
```

Rules:

- **The evaluating node resolves it, with its own credentials.** `pg://warehouse` names a connection the node knows; the expression never carries a secret, because the expression is the shareable audit artifact (Part A §1.2). The spec's delegated-credential token can only target `abc_node` or `http_source` (Part H §31.4), so node-owned credentials are the only way a non-HTTP scheme gets any.
- **The adapter types the result** (§8): an S3 object carries its stored `Content-Type`; a table or query returns `application/json` rows, or `text/csv` when asked with `;accept`. Rows are a collection, so `pg://warehouse/analytics.monthly_sales*(row)!'summarise $item'` iterates them (Part B §5.3).
- **Advertised, or refused.** The node lists its schemes in the capabilities document; an unlisted scheme is `unsupported_mode`, permanent (Part B §3.5). Access control and consent apply as for any source (Part B §5.6.4). Such a source is not url4-aware: the envelope marks it `abc_aware: false` and attribution stops at it (Part G §26.4), the same as for any plain HTTP read.
- **Hexagonal.** A scheme adapter is an `IOLayer` adapter registered by scheme; the core never imports it. `url4 serve` gains a `[schemes]` table beside `[data]`.

What a scheme's path means (bucket and key; database, schema and table; file and table) is the adapter's contract, not the grammar's. The grammar only sees `scheme://authority/path`.

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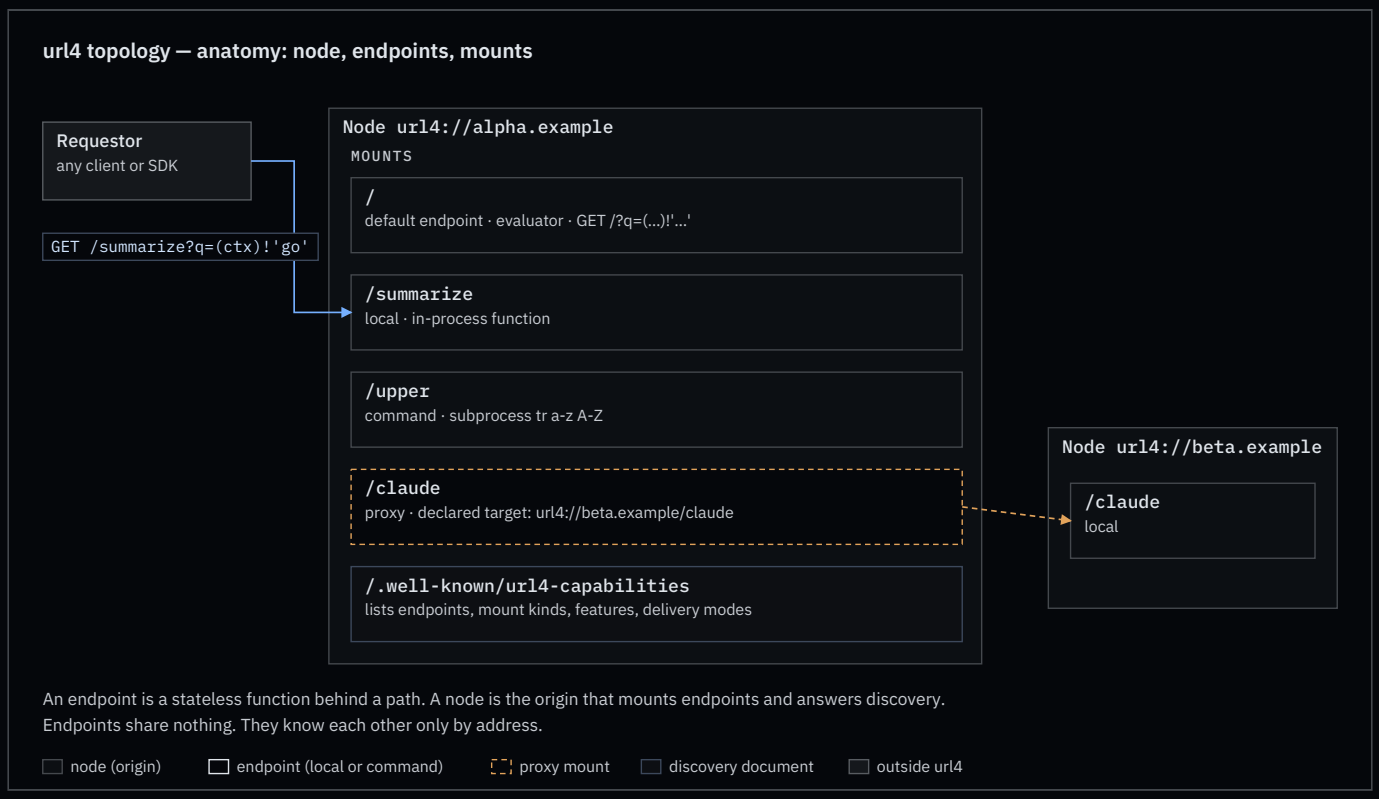
3. Endpoint contract

An endpoint is a function. That is the whole idea.



An endpoint:

- **MUST** answer `GET <path>?q=<expression>` and evaluate the expression. Every endpoint evaluates url4; an endpoint may walk a DAG before it reaches its own intent processor. There is one grade of endpoint, not two.
- **MUST** be stateless per invocation. It keeps nothing between calls. Its own data is reached through `@` (Part B §5.6) and lives behind it, not in it. The one exception the spec defines is an agent session (Part G §28): the coordinating node holds the session and its transcript for the session's life; participant endpoints stay stateless because the transcript travels with every turn. `;coord=` selects a coordination mode (`session`, `debate`, `roundrobin`, `freeform`); the session id is endpoint-issued (`Session-Id` header), not requestor-supplied.

- **MUST** return the envelope (Part D §17) and state the delivery mode it actually used (Part C §11.4).
- **MUST NOT** resolve `@` inside a sub-expression addressed to another node (Part B §5.6.3.1).
- **MUST** answer sync. **SHOULD** offer stream (SSE), WebSocket, and async, and advertise which (§5, §6). When asked for more than it has, it answers with the richest mode it does have; that is never an error.
- **MAY** be mounted on any node, or be its own origin. A Lambda-style function with its own URL is a node with one endpoint.
- Speaks GET. A WebSocket, when offered, is the same GET with `Upgrade: websocket` (§6). Other verbs are 405, except the ones this document adds: `DELETE` on a run handle (§6) and, if adopted, `OPTIONS` (§5).

Endpoints share nothing. An endpoint knows another endpoint only by address, and only because the expression named it.

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4. Node contract

A node is an origin. It routes; it does not evaluate.

- Mounts one or more endpoints at paths. `/` is the default endpoint.
- Mount kinds: `local` (a function in the process), `command` (a subprocess; doctrine N4), `proxy` (forwards to a declared target on another node). These are the spec's intent-processor types under our names: `internal` / `function`, `code`, and `abc_delegate` (Part G §27.3). A proxied relative call is exactly the spec's delegation: the caller has already resolved the context, so what crosses the wire is materialised sources plus the intent, never a raw sub-expression. A `command` mount is code execution and inherits the pinning and signing rules of Part H §36.
- Declares proxy mounts with their targets in its capabilities document. Attribution and consumer disclosure (Part H §29.2.1) need the real target, so proxies are never hidden; a source may also cap redistribution depth or name permitted consumers (Part H §29.2.2), and a proxy target is a consumer.
- Serves discovery (§5) and the version alias `/v1` (§2).
- A local SDK process that mounts remote endpoints under local names is a node. That is how "run the ensemble on my laptop, but call `/claude` as if it were mine" works.

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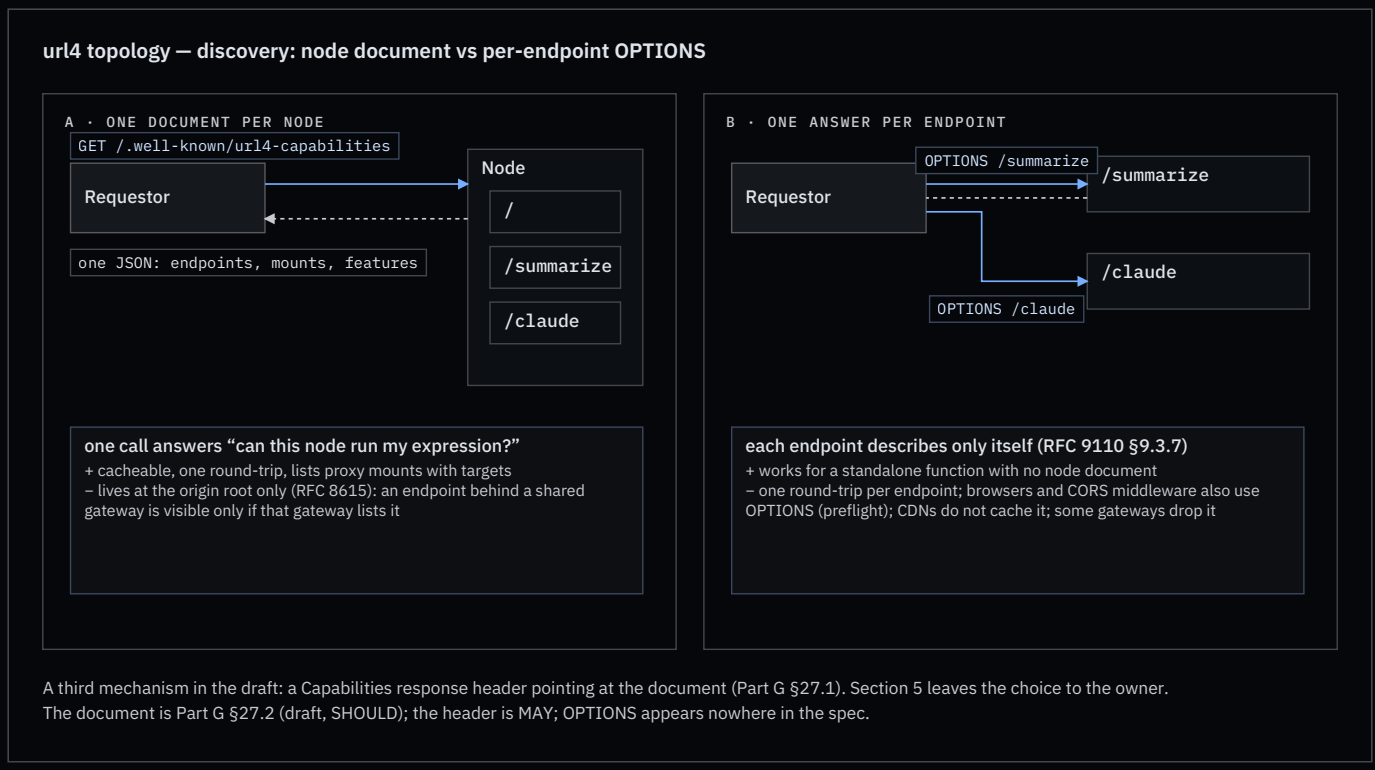
5. Discovery — the open decision

Three ways for a requestor to learn what a node can do. The draft Part G specifies the first two: a capabilities document at `/.well-known/abc-capabilities` (SHOULD) and a `Capabilities` response header that MAY point to

it (Part G §27.1). OPTIONS is ours; it appears nowhere in the spec. The spec also reaches for `.well-known` for the policy registry (`/.well-known/abc-policy` , Part H §30.2), so two well-known documents will coexist.



A third mechanism in the draft: a Capabilities response header pointing at the document (Part G §27.1). Section 5 leaves the choice to the owner. The document is Part G §27.2 (draft, SHOULD); the header is MAY; OPTIONS appears nowhere in the spec.

	A · NODE DOCUMENT `GET /.well-known/url4-capabilities`	B · PER ENDPOINT `OPTIONS /<endpoint>`	C · `Capabilities` RESPONSE HEADER
“Can this node run my expression?”	one call, whole answer	one call per endpoint named in the expression	pointer only; still one fetch of A
Standalone function with its own origin	works: the document sits at that origin’s root	works	works
Endpoint behind a shared, non-url4 gateway	invisible unless the gateway lists it (RFC 8615: root of the origin only)	works: the endpoint answers for itself	works: any ordinary response can carry the pointer
Caching	ordinary GET: `ETag` , CDN, browser cache	not cached by CDNs or browsers	rides on the cached response
Browsers and CORS	plain GET	preflight uses the same verb; CORS middleware often answers first (RFC 9110 §9.3.7 allows a body, browsers ignore it)	plain header
Gateways and serverless platforms	fine	some strip or auto-answer OPTIONS	fine
Proxy mount targets (attribution)	listed in one place	per endpoint	via A
Spec status	Part G §27.1, SHOULD; schema Part G §27.2	not in the spec	Part G §27.1, MAY

Both share one hazard: the Engine’s JWT header is called `URL4-Capability` . The document is “capabilities”. Appendix B proposes renaming the header.

Capabilities document: the spec's shape, plus our additions. Part G §27.2 already defines the top level: `abc_version`, `node` (one canonical URI: our node), `self_ref_support`, `collections[]` (`id`, `path`, `description`, `element_schema`, `formats`, …), `behaviors`, `intent_processors[]` (`id`, `type` ∈ `internal` | `abc_delegate` | `http_delegate` | `code` | `function`, `endpoint`, `capabilities`), `processor_policy`, `mode_support`, `agent_profiles`. Content types come from Part F §25.4 as `consumes` / `produces`. Sub-paths are read as collections (Part G §27.4). Our additions, marked, are what makes a processor addressable as an endpoint:

```
{
  "abc_version": 1,
  "node": "url4://alpha.example",
  "default_processor": "summarize",                        // ours
  "intent_processors": [
    { "id": "summarize", "type": "internal", "path": "/summarize",   // path: ours
      "consumes": ["text/plain"], "produces": ["text/plain"],
      "delivery": ["sync", "stream"] },                                  // delivery: ours
    { "id": "upper",     "type": "code",     "path": "/upper" },
    { "id": "claude",    "type": "abc_delegate", "path": "/claude",
      "endpoint": "url4://beta.example/claude" }
  ],
  "schemes": ["s3", "pg", "sqlite"],                         // ours (§2)
  "collections": [ { "id": "science", "path": "/science", "self_ref_eligible": true } ],
  "self_ref_support": true
}
```

One point to settle with Kevin: the spec's document describes one node whose processors are addressed by `id` and whose sub-paths are collections; our endpoints need a path each. `intent_processors[].path` is the smallest change that gives both.

> **Owner decision (open).** Pick one:
>
> ☐ **A only** — node document is a MUST; nothing per endpoint.
>
> ☐ **B only** — OPTIONS per endpoint is a MUST; no node document.
>
> ☐ **A MUST + C SHOULD** — the spec's pair, one level stronger than its SHOULD/MAY.
>
> ☐ **A MUST + B SHOULD** — the document indexes the node; an endpoint may also answer OPTIONS with its own entry.
>
> Recommendation, revised after reading Part G: **A MUST + C SHOULD**, B dropped. The `Capabilities` header solves the gateway case that motivated OPTIONS, and it is already in the draft.

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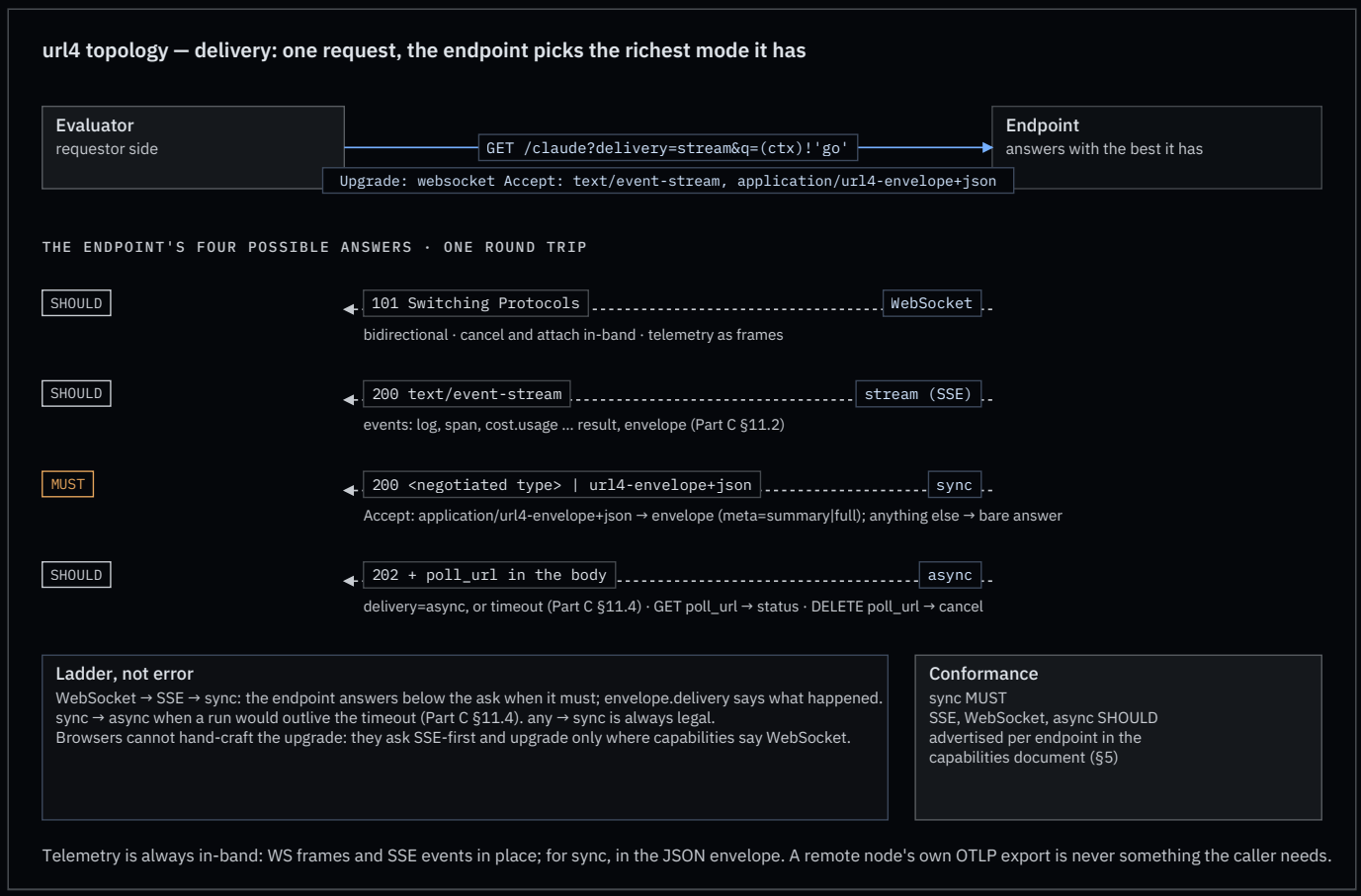
6. Delivery and transport

The spec defines three delivery modes and a fallback rule (Part C §11), and a graceful-degradation rule that an endpoint MUST degrade rather than fail and MUST report what it did (Part C §10.2). We adopt them, add WebSocket as

a fourth answer, and make the whole ladder one request. Part I §43 question 26 rules `ws://` sources out of scope; that is a different question from a WebSocket answer on the endpoint's own GET.



One request, the endpoint picks. The evaluator sends the richest ask it can, once:

```
GET /claude?delivery=stream&q=(ctx)!'go'
Upgrade: websocket
Accept: text/event-stream, application/json
```

The endpoint answers with the best it supports. RFC 6455 lets a WebSocket handshake ride an ordinary GET; RFC 9110 lets a server ignore `Upgrade` and answer normally. So one round trip covers the whole ladder:

ENDPOINT ANSWERS	MEANING	TELEMETRY TRAVELS AS
`101 Switching Protocols`	**WebSocket**: bidirectional; cancel and attach in-band	frames, same names as the SSE events
`200 text/event-stream`	**stream**: SSE on the same GET (Part C §11.2)	SSE events, then `result`, then `envelope`
`200 text/plain` or `application/json`	**sync**: bare answer, or the envelope, by `Accept`	envelope `meta` (below)
`202` + `poll_url`	**async**: `delivery=async`; the 202 body carries `job_id`, `rid`, `poll_url` (Part C §12.5); `GET` it for status, `DELETE` it to cancel	on the handle

Rules:

- **sync is the only MUST.** SSE, WebSocket and async are SHOULD, advertised per endpoint in the capabilities document (§5). A serverless function that offers only sync and SSE is conformant.

- **Ladder, not error.** WebSocket → SSE → sync. An endpoint answering below the ask is normal; the envelope's `delivery` field says what happened (Part C §11.4), and the degradation is reported like any other (Part C §10.2.2). The spec's ladder has two edges, `stream → sync` and `sync → async`; the WebSocket rung and `any → sync` are ours.
- **Async is the spec's third `delivery` value.** It is requested with `delivery=async` (Part C §9.1), or entered by the endpoint when a sync run would outlive the timeout (Part C §11.4 `sync → async`). The Engine's `Prefer: respond-async` becomes an alias. The run handle is the `poll_url` field of the 202 body; a `Location` header mirroring it is our addition.
- **Browsers go SSE-first.** A browser cannot attach `Accept` to a WebSocket handshake, so a browser evaluator asks for SSE and upgrades only where the capabilities document says WebSocket is offered.
- **The Engine's product session** keeps its WebSocket at `/ws`. It is one instance of the WebSocket rung, not a separate protocol.
- **Telemetry is always in-band.** Whatever the mode, the caller receives logs, spans and cost on the same connection or in the envelope. A remote node's own OTLP export is its business, never something the caller depends on (§7).

Telemetry per mode. WebSocket and SSE carry it in place: an SSE body is a sequence of named events (`event: log`, `event: span`, `event: cost.usage`, then `event: result`, `event: envelope`), in order, on the one response. The spec already builds on this (Part C §12.5 is an SSE event catalog); our three signals are three more names. What SSE lacks against WebSocket is only the return path: no in-band cancel or attach.

Sync has no room for that: the body is the answer. The spec always returns the JSON envelope and uses `Accept` to negotiate the result's content type (Part C §9.1, Part F §25.2). We add one wrapper switch on top, named so it cannot collide with that:

KNOB	GOVERNS	VALUES
`Accept` header	the wrapper, by one dedicated type	`application/url4-envelope+json`: the envelope (Part D §17). Anything else: the bare answer in the negotiated content type, no logs, no telemetry, what `url4 serve` returns today. Our addition; the spec has no bare mode
`meta` param	how much the envelope carries	`none`: result and status. `summary`: counts, latency, cost. `full`: per-source detail, nested child envelopes (Part D §17.3), and a `telemetry` block with logs and spans. The `telemetry` block is ours; per the Hybrid Rule it is present and `null` at `summary` (Part D §17.2.2)
`fmt` param	the shape of the answer content	`text`, `markdown`, `json` (Part C §9.1); orthogonal, a JSON envelope can wrap a markdown answer

An envelope answer with no `meta` gets `summary` (the spec's default is `none`, Part C §9.1), so asking for the envelope always buys the cheap insight; `meta=full` is the explicit debug switch. What an endpoint exposes is its decision (Part D §18.4): a production endpoint may answer `meta=full` with the `telemetry` block redacted, but it MUST keep the structure and use `null` or `"redacted"`, never drop fields silently. A development endpoint hands over everything. Same envelope, same evaluator code, different policy.

Cancel. The spec calls cancellation a gap and offers two shapes, `DELETE <poll_url>` or a `cancel=<rid>` parameter (Part C §16.2). We take the first: over WebSocket, cancel is in-band; otherwise `DELETE` on the `poll_url`, which the Engine already does. New terminal state `cancelled`; SSE event `request.cancelled`; partial result returned when quorum was already met, as §16.2 asks. Note for Kevin: `status: "cancelled"` is not yet in the status enum of Part D §17.4.

7. Telemetry

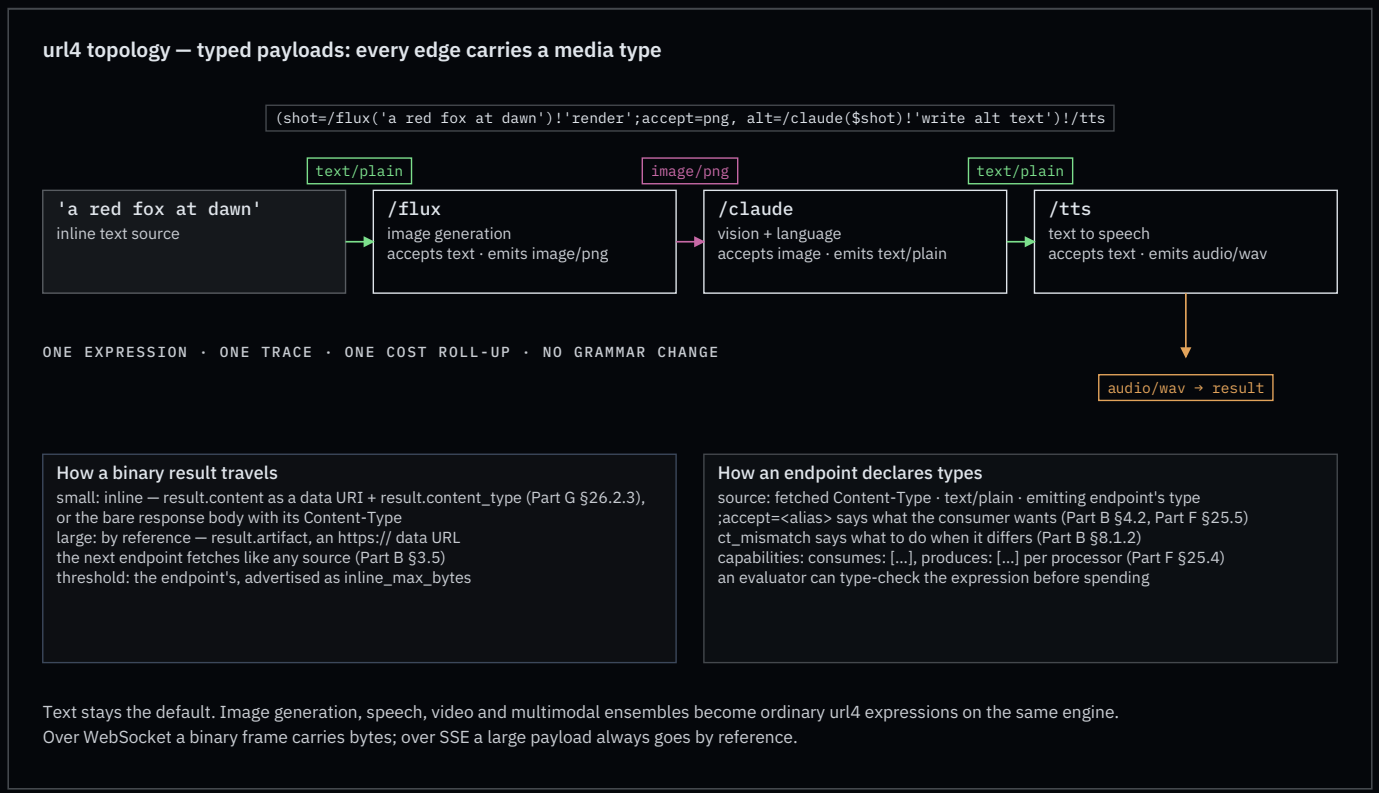
Three signals, one trace, as in the doctrine skill. What changes is where a one-shot GET puts them.

- **sync**: in the body, only when the caller asked for the envelope (`Accept: application/url4-envelope+json` , §6). `meta` sets the level (`summary` by default, `full` adds logs and spans; Part D §17.2). At `meta=full` a child's envelope nests in `source.envelope` (Part D §17.3). Each endpoint aggregates only what it saw itself (Part D §18.1). A bare `text/plain` answer carries nothing.
- **stream** and **WebSocket**: as events: logs, spans, `cost.usage` (self and subtree), then `result` , then `envelope` . Same names in both.
- **durable**: OTLP export to the trace backend. There is no separate "Enclave" store; the exporter superseded it.
- **What the spec already carries**: 21 SSE event types (Part C §12.5), `meta.total_cost` and `budgets_spent` in the envelope (Part D §17.1.2.8, Part E §24), and an endpoint-local audit log (Part H §34). Our logs, spans and `cost.usage` events are additions on top of that catalog, not replacements; cost stays a separate event, and `total_cost` is its roll-up. Every streamed event with content already MUST carry `content_type` (Part C §12.5), which is what §8 builds on.

This closes doctrine item F4. **Idempotency:** a url4 GET is idempotent in the HTTP sense (RFC 9110 §9.2.2): repeating it has no extra server-side effect. Results need not be identical; Part C §15.1 conflates the two (Appendix A). One real exception the spec lists: a repeat that would exceed a consumed budget fails with `budget_exceeded` (Part C §15.2).

8. Typed payloads: an endpoint is a universal inference processor

We have treated an endpoint as text in, text out. That is a habit, not a spec rule. The spec defines a source as "a URI, text, or nested expression" and an intent processor as whatever executes the intent on behalf of the endpoint (Part A §1.4). Nothing says the payload is a string.



Proposal. Every edge in an expression carries a **typed payload**, the way a ComfyUI edge carries an image, a latent or a mask. The type system is the media type: `text/plain` , `image/png` , `audio/wav` , `video/mp4` , `application/json` for embeddings and structured data. Text stays the default and the most common type. No grammar change.

```
(shot=/flux('a red fox at dawn')!'render';accept=png,
 alt=/claude($shot)!'write alt text')!/tts
```

Edges: text → image → text → audio. Three endpoints, three processors, one expression, one trace, one cost roll-up.

What the spec already gives us

- `;accept` is a source-level execution annotation (Part B §4.2) that takes a **short alias** (`json`, `csv`, `markdown`, …), never a MIME type, because `/` would read as a relative URI (Part F §25.5). The registry has no image, audio or video rows yet; `png`, `jpeg`, `wav`, `mp4` are proposed in Appendix A. `ct_mismatch` has four values, `fail | ignore | transform_ignore | transform_fail`, default `ignore`, and only applies when `;accept` was declared (Part G §26.3). The spec calls input negotiation out of scope and treats an unusable format as an intent-execution failure, not a source failure (Part F §25.10).
- The spec already has a weighted `image/png` source feeding a text intent (Part I §41.20), and Part G §26.2.3 already says how binary content travels in LLM mode: a data URI, `data:<mediatype>;base64,…`, with raw bytes for RDS intents. The SDK carries a media type on every fetch (`FetchRequest.media_type`) and parses collections by declared type (Part B §5.3.7). The Engine already forks results by size: inline, spilled to a content-addressed artifact, or refused above a hard cap.

Three rules the spec does not yet give (Part F, §25)

1. **A source declares its type.** A URI source has the `Content-Type` it was fetched with; absent that, the endpoint infers from the path or sniffs, and falls back to `application/octet-stream` (Part G §26.2). Inline text is `text/plain`. A nested expression has the `result.content_type` its endpoint emitted (Part D §17.1.2.1). `;accept` says what the consumer wants; `ct_mismatch` says what to do when the two differ.
2. **A binary result travels inline when small, by reference when large.** Bare sync: the response body is the bytes, with its `Content-Type` (Part F §25.5 permits raw binary outside the envelope). Envelope: `result.content` carries a small payload as a data URI with `result.content_type` set, the spec's own form; a large payload becomes `result.artifact`, an `https://` URL that the next endpoint fetches as plain data (Part B §3.5). The inline threshold is the endpoint's, advertised in its capabilities. Over WebSocket a binary frame carries the bytes; over SSE a large payload always goes by reference. Two cautions from the drafts: this is not truncation, which Part G §26.2.1 forbids; and an artifact URL is a redistribution, so it inherits the source's flow constraints and depth cap (Part H §29.2.2).
3. **A processor advertises what it accepts and emits.** The spec's names are `consumes` and `produces` (Part F §25.4), per processor in the capabilities document. An evaluator can then type-check an expression before it spends anything, which is what the plan question in §10 builds on.

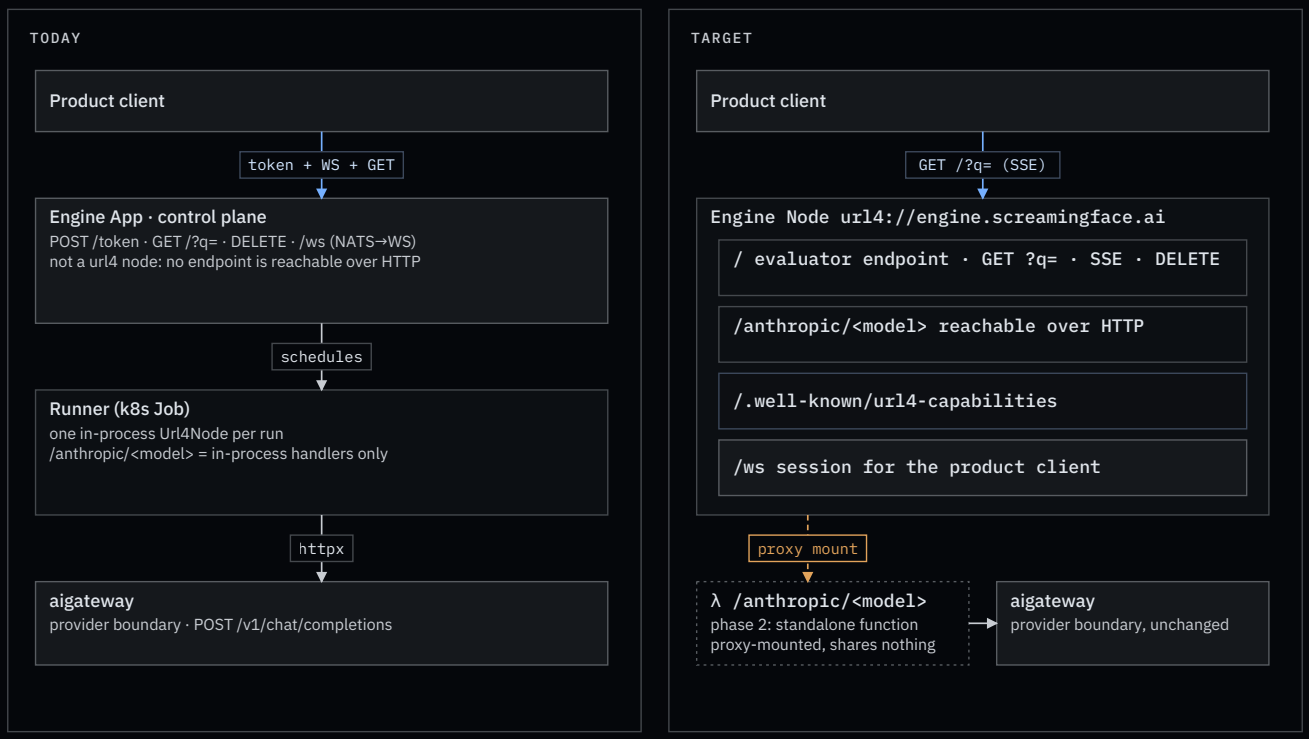
```
{ "id": "flux",   "type": "internal", "path": "/flux",
  "consumes": ["text/plain"], "produces": ["image/png"], "inline_max_bytes": 262144 },
{ "id": "claude", "type": "internal", "path": "/claude",
  "consumes": ["text/plain", "image/png", "image/jpeg"], "produces": ["text/plain"] },
{ "id": "tts",    "type": "internal", "path": "/tts",
  "consumes": ["text/plain"], "produces": ["audio/wav"] }
```

Why it matters. Image generation, text-to-speech, speech-to-text, video evaluation and multimodal ensembles all become ordinary url4 expressions on the same engine, with the same telemetry, the same cost accounting and the same attribution. aigateway stays the provider boundary; it already fronts image and audio providers. The Engine's artifact store is rule 2 as built.

Open. A media type for embeddings (`application/json` with a profile, or a vendor type); how attribution weights and `tokens=` budgets apply to non-text sources (Part E is silent; no `bytes=` or `seconds=` budget key exists). Settled by the draft: `fmt` stays a structure hint and a future version MAY retire it in favour of `Accept` (Part F §25.8).

9. The Engine as a node

url4 topology — the Engine today and as a url4 node



The Runner becomes the evaluator process the / endpoint spawns per run. The App keeps /ws for the product client and gains the endpoint surface.
Serverless posture: endpoints are functions; the node is the only thing that needs an origin.

	TODAY	TARGET
App	control plane: `POST /token`, `GET /?q=`, `DELETE`, `/ws` bridged from NATS	a **node**: `/` evaluator endpoint (`GET ?q=` with WebSocket, SSE, sync and async answers, `DELETE`), model endpoints over HTTP, capabilities, `/ws` kept for the product client
Runner	k8s Job with one in-process `Url4Node`; model routes are in-process handlers	the evaluator process the `/` endpoint spawns per run
Model endpoints	`/anthropic/<model>` reachable only inside the Runner	reachable over HTTP on the node; later each one a standalone function, proxy-mounted
aigateway	provider boundary	unchanged
Endpoint-to-endpoint	unused SDK code	the normal path

Phases. (1) Node surface on the App: discovery per §5, `GET /?q=` answering SSE and sync (WebSocket via `Upgrade` where the App already has it), `DELETE`. (2) Model endpoints over HTTP. (3) Standalone model functions, proxy-mounted, listed in the capabilities document.

Serverless posture. An endpoint is a function; it needs no origin of its own. Only the node needs one. The Runner is already function-shaped. The App is not: it holds an audience count in memory, runs two perpetual tasks, and gates runs on an attached WebSocket. Those belong to the product session, not to the endpoint surface, and stay in the App.

Appendix C — where the spec lives

- Parts A and B, v0.5 DRAFT (2026-07-10): `secondbrain/kevin-mcdonough/docs/adrs/URL4-Spec-A.md`, `URL4-Spec-B.md`.

- Parts C–I: draft markdown on the `url4-refactor` **branch** of `OpenMined/screamingface-design`, `kevin-mcdonough/docs/adrs/refactor/URL4-Spec-{C…I}.md`, cut 2026-04-28 from the v0.4 text, not yet reviewed, never merged to `main` (where the site shows "Not yet written" stubs). Cited here as "Part X §N". The same branch holds the v0.4 monolith (commit `f28608a`); the v0.2 monolith this document was first written from is `8a052dc`, and its numbering diverges from §21 on.
- Public docs: `public-docs/src/pages/learn/Url4Page.vue`. They use "fusion" and "typed DAG"; the spec uses neither.
- Doctrine: `.claude/skills/url4-engine/SKILL.md`, updated with this document.

Appendix D — Vocabulary

One line each: what it is, why it matters, where the spec grounds it. Terms marked ours are proposals in this document.

Language

TERM	DEFINITION	ANCHOR
Expression	`(sources)!intent`: given this data, do this. The atomic unit of work and the whole composition; sources may be expressions, so expressions nest into a tree.	Part A §1.4, Part B §2
Source	One input: inline text, a URI, or a nested expression. Carries attribution annotations (`name:weight:budget`) and execution annotations (`;t`, `;retry`, `;accept`).	Part B §4
Intent	The right side of `!`: a prompt, a code pointer, a relative or remote URI, or a computed expression.	Part B §6
Intent processor	What turns resolved context plus intent into a result: a model call, a script, a command, or a delegate on another node.	Part A §1.4, Part G §27.3
Holdings, `@`	The endpoint's own data via the self-reference token; `@alice` names a principal's data under access control and consent.	Part B §5.6
Collection	A source that parses into rows; `*` iterates it. A sub-path after an endpoint also selects a collection.	Part B §5.3, Part G §27.4

Topology

TERM	DEFINITION	ANCHOR
Endpoint	A stateless function at a path (`/claude`). Answers `GET <path>?q=<expr>`, evaluates it, returns a result. One grade: every endpoint evaluates url4. The spec's word.	§3, Part A §1.4
Node	The origin that serves a set of endpoints, `/` being the default, publishes discovery, owns credentials and outbound traffic. Every requester running an evaluator is one. The spec's word.	§4, Part A §1.4
Mount	The binding of a path to an endpoint implementation: `local`, `command`, `proxy`. The spec's processor types `internal` / `function`, `code`, `abc_delegate`.	§4, Part G §27.3
Evaluator	The code that runs an expression: resolves sources, fans out, reduces, runs the intent. Every endpoint contains one.	§1
Requestor	Whoever submits an expression.	Part A §1.4
Request tree	The nodes and endpoints one expression touches; strictly a tree. No plan object, no swarm.	Part H §29.1

TERM	DEFINITION	ANCHOR
Capabilities document	JSON a node publishes at `/.well-known/url4-capabilities`: processors, collections, delivery modes, schemes. A `Capabilities` header may point to it.	Part G §27.1, §27.2
Scheme adapter (ours)	A node-mounted reader for a non-url4 scheme (`s3://`, `pg://`, `sqlite://`) with the node's own credentials, typing its result, advertised in capabilities.	§2, Part B §3.5

Transport

TERM	DEFINITION	ANCHOR
Delivery mode	How the answer returns: `sync`, `stream` (SSE on the same GET), `async` (202 + `poll_url`), and our WebSocket rung (`101` on the same GET). Sync is the only MUST.	§6, Part C §11
Ladder (ours)	The endpoint answers with the richest mode it supports, WebSocket → SSE → sync, in one round trip; answering below the ask is a reported degradation.	§6, Part C §10.2
Envelope	The JSON wrapper: `result`, `status`, `delivery`, `sources[]`, `meta`. Requested with `Accept: application/url4-envelope+json`; otherwise the bare result.	Part D §17
`meta`	Envelope depth: `none`, `summary` (counts, latency, cost), `full` (per-source detail, nested envelopes, telemetry).	Part D §17.2
Run handle, `poll_url`	The address of an async run: `GET` for status, `DELETE` to cancel.	Part C §12.5, §16
`rid`	The request id; fresh per child request, recorded by the parent, bound into tokens.	Part C §9.1
Telemetry signals	Logs, spans (tokens live here), and `cost.usage` events (money lives here). In-band in every mode; OTLP is the durable copy.	§7

Data

TERM	DEFINITION	ANCHOR
Typed payload (ours)	Every edge carries a value named by its media type; text is the default. Sources declare by `Content-Type`, results by `result.content_type`.	§8
Artifact (ours)	A large result returned as an `https://` URL the next endpoint fetches as data, above an endpoint-declared `inline_max_bytes`; inherits flow constraints.	§8
`;accept`, `ct_mismatch`	A source's wanted format, as a short alias, and what to do when the fetched type differs: `fail`, `ignore`, `transform_ignore`, `transform_fail`.	Part F §25.6, Part G §26.3

Trust

TERM	DEFINITION	ANCHOR
Run session (ours)	A node-issued capability scoped to one run (`rid`, tree, purpose, expiry, identity) that endpoints carry instead of credentials.	§10, §11
Inter-node token	The requestor's credential for one node, encrypted to that node's key, bound to `rid`, a timestamp (≤ 300 s) and the destination; intermediaries forward what they cannot decrypt.	Part H §31

TERM	DEFINITION	ANCHOR
Egress (ours)	The node component that performs every outbound request for its endpoints: policy, disclosure, cache, budgets, credentials, token forwarding.	§11
Policy registry	The out-of-band service a source exposes to state its terms; consulted before resolution.	Part H §30
Flow constraints	A source's limits on where its output may travel: permitted or denied consumers, redistribution depth.	Part H §29.2.2
Attribution	The per-source influence score the envelope reports; weights and budgets shape it.	Part E

Execution

TERM	DEFINITION	ANCHOR
Quorum, trigger	Quorum: how many sources must succeed before the intent may run. Trigger: the terminal-source count at which the endpoint decides whether to produce a result.	Part C §12
Degradation	An endpoint must fall back rather than fail when it cannot honour a request, and must report it.	Part C §10.2
Agent session	A multi-turn interaction among `mode=agent` sources coordinated by the resolving endpoint, which holds the transcript; `coord=` selects the mode.	Part G §28
Plan, dry run (ours)	A node endpoint that resolves and type-checks an expression, consults policy and budgets, and returns the envelope with no result. Open question.	§10
Idempotent	A url4 GET has no extra server-side effect on repeat (RFC 9110); results need not be identical. Disputed wording in the spec.	Part C §15